SABIL

Solar Access Builds Independent Lives

A home solar kit whose controller makes the battery last, sold per kWh to families in Maiduguri who lose the grid for months.

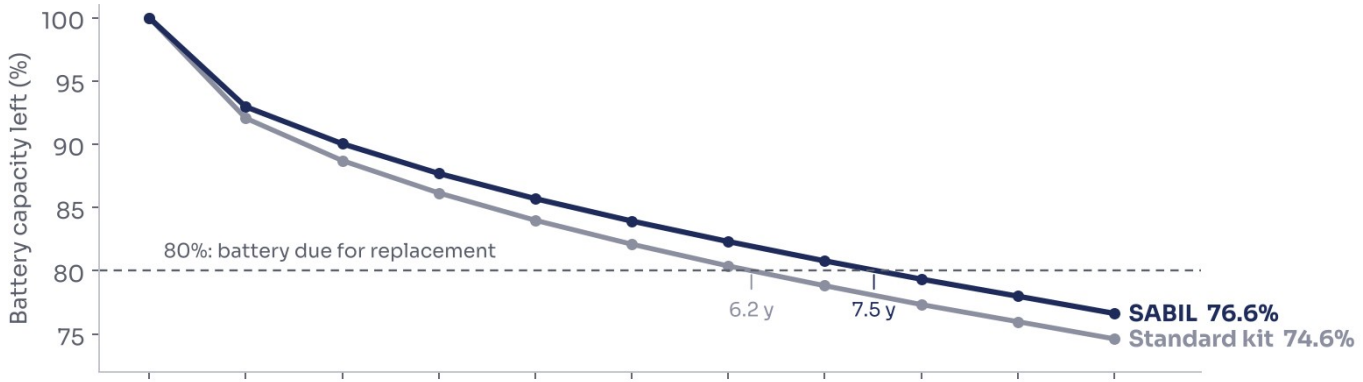
Brief	13. Power for a city the grid can't protect
Track	4. Clean Energy Solutions. Setting A, residential: homes in Maiduguri
Team	Youcef Toukal, materials engineer: battery ageing, kit sizing, cost model
	Younes Toukal, AI engineer: weather data, simulation, controller, forecasts, interface

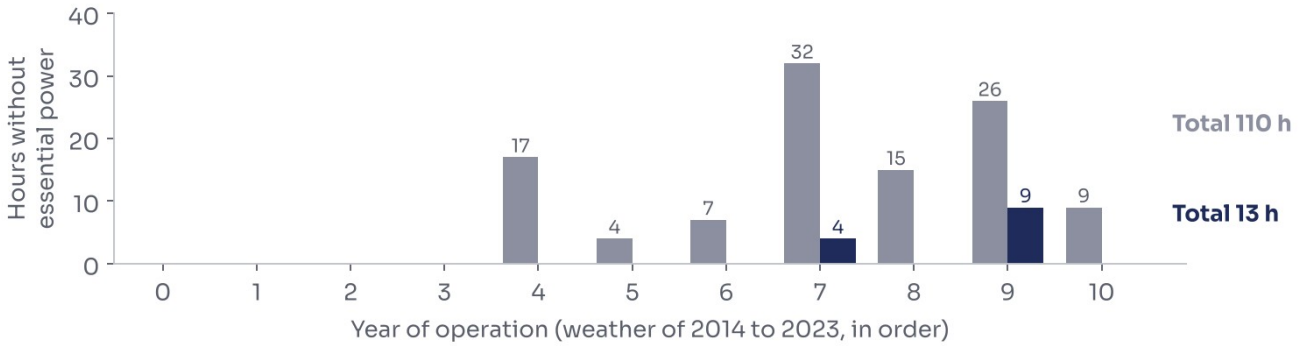
Global Energy Hackathon 2026. Scored on impact 25%, innovation 25%, sustainability 25%, feasibility 15%, scalability 10%: one slide each, in that order.

When the line is cut, a standard solar kit loses 110 hours of essential power in 10 years. SABIL loses 13.

Who is affected

Maiduguri runs on one 330 kV line through insurgent-held land. In 2021 it was blown up again and again; the city spent about ten months without grid power. Families fall back on petrol generators, at $1.07 a litre (₦1,430) in Borno, Oct 2026.





> *"One ice seller ran 4 of his 40 freezers and cut his staff from 300 to 50."*
> Brief 13, reporting on the 2021 blackout

110 → 13 h essential hours lost in 10 years

6.2 → 7.5 y battery life to 80% capacity

212–849 L fuel avoided per home per year

Simulation: 400 Wp + 1.92 kWh kit, 5-person home (1.16 kWh/day), NASA POWER hourly weather 2014–2023, grid down. Ageing: Naumann et al. 2018, 2020.

In Maiduguri's heat, the battery, not the panel, decides how long a solar kit keeps the lights on.

Today

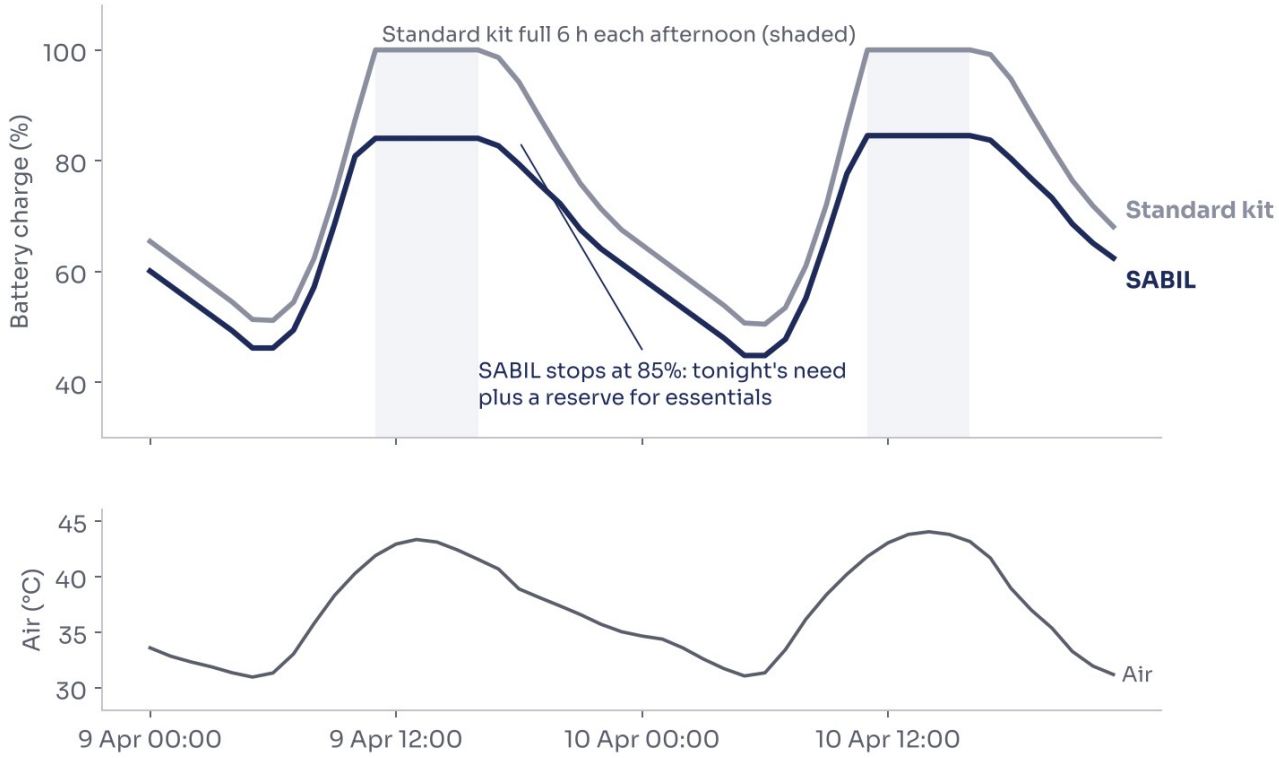
Home solar kits charge to 100% and sit full all afternoon at 35 to 44 °C. Lithium batteries wear fastest when hot and full, so essentials start failing from year 4.

Our approach

A controller that knows how batteries age: on hot days with a sunny tomorrow it charges only what tonight needs, keeps a reserve for essentials and serves them first. Families pay per kWh; the operator guarantees the battery.

Root cause we address

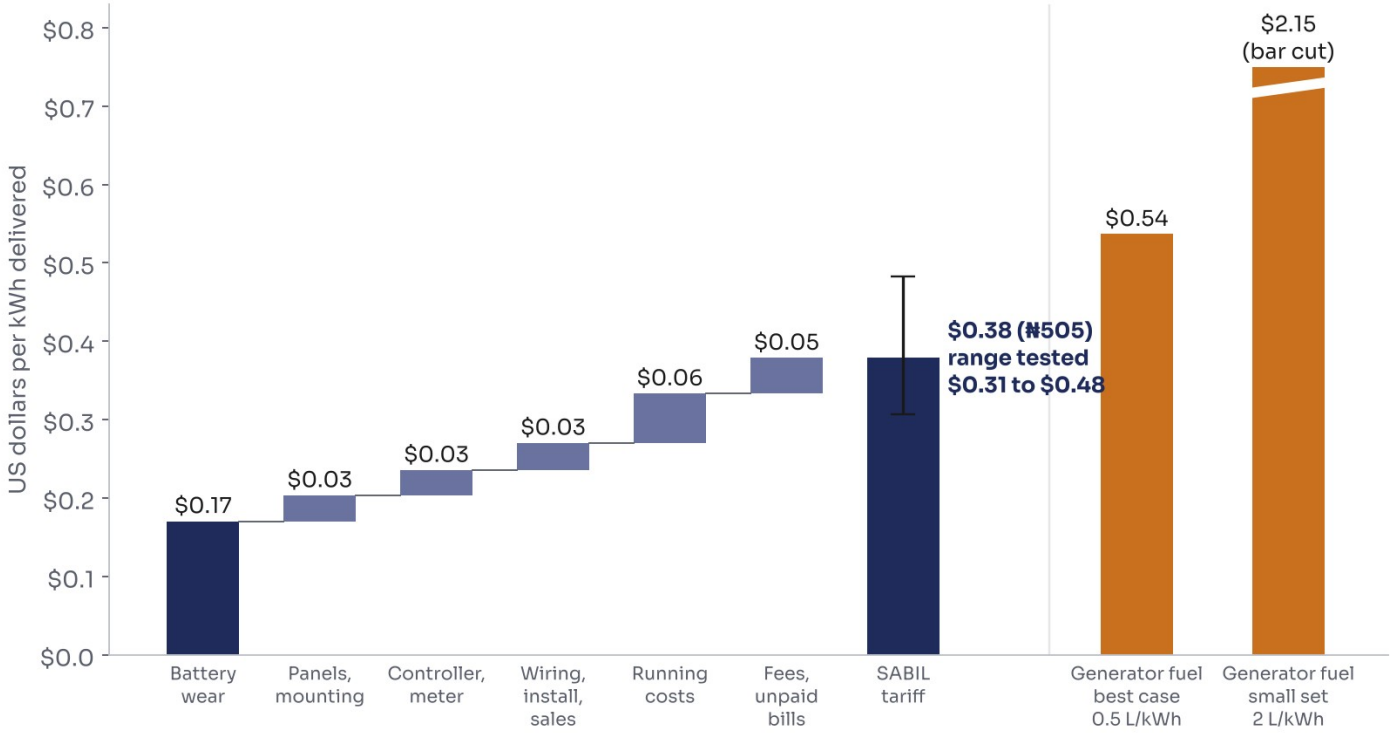
Brief causes 4 and 5: generators are the fallback, essential loads are not protected.



1. **Forecast:** tomorrow's sun from today's, or a learned model (12 compared).
2. **Plan:** tonight's energy plus an essentials reserve sets today's charge ceiling.
3. **Run:** essentials first, flexible loads to sunny hours, top-ups by phone.

Battery charge and air temperature, 9-10 April 2016, hourly simulation on NASA POWER weather. Ageing model verified against SimSES (TU Munich) to 4.5×10^{-14}.

A SABIL kWh costs $0.38 (₦505), below even the best-case fuel cost of a petrol generator ($0.54).



Break-even tariff built up from its parts (kit over 20 years at 12% real), against generator fuel alone at the Borno pump price. Bar for SABIL: range over all assumptions tested.

Cost to the user

$4.06 a month for lights, phones and fans; $13.40 with a fridge. Minimum wage: $53 (₦70,000). Nothing up front.

Compared with today

The same kWh from a petrol generator: $0.54 to $2.15 in fuel alone, or $19 to $76 a month.

Payback

The controller's $12 is repaid in 1.3 years by slower battery wear.

Who pays

Families per kWh by mobile money; the operator owns the kit. A DARES grant (~$84) cuts the tariff to $0.35/kWh.

How it pays for itself

The tariff covers the kit, running costs, fees and 10% unpaid bills. No ongoing subsidy.

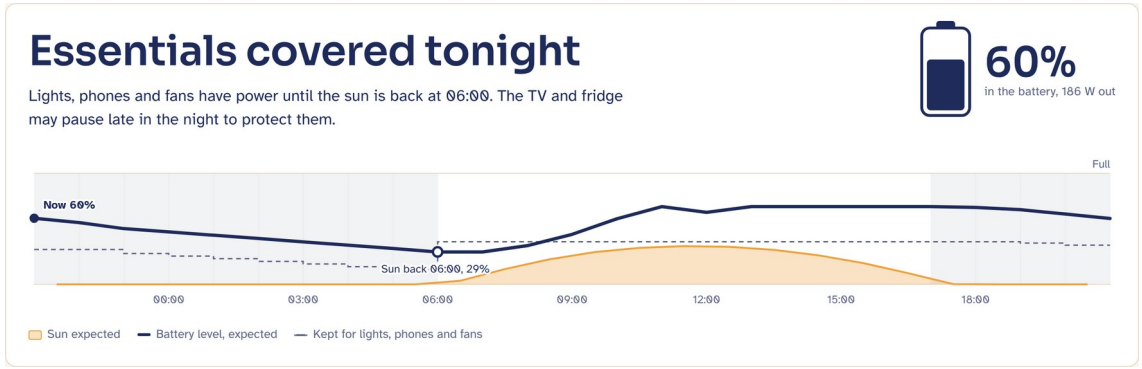
Who keeps it running

Local agents install and service; remote monitoring flags heat, dust and battery health.

Environment. 212 to 849 L of fuel a home a year; 542 fewer battery modules (9.8 t) per 1,000 homes over 20 years.

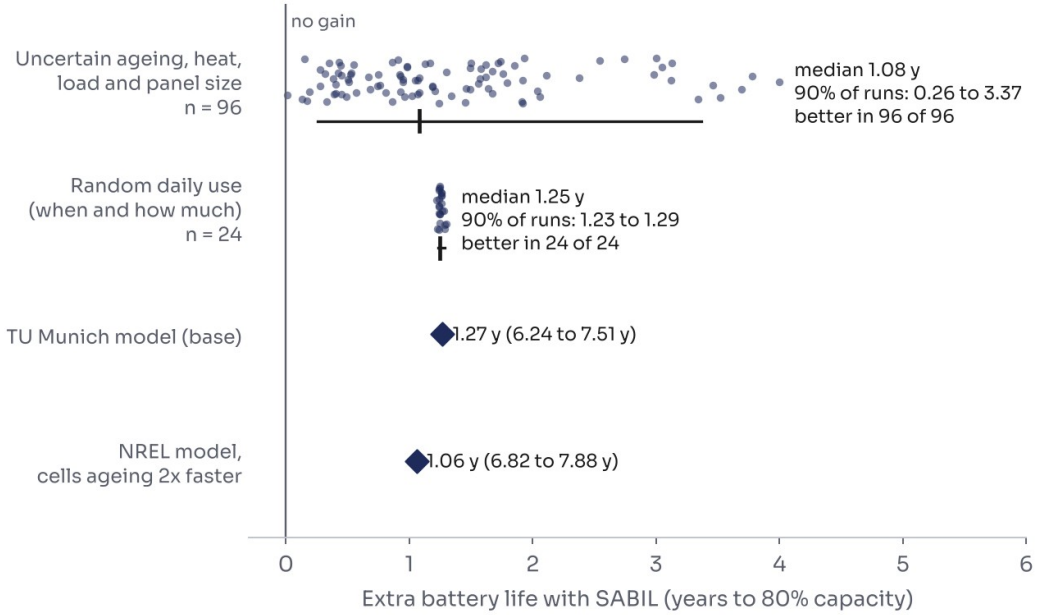
src/cost_model.py; params/costs.json: NNPC Oct 2026, NBS, CBN ₦1,332/USD, PVPRO Sep 2026, BNEF Dec 2025, World Bank DARES 2023, A2EI 2022.

SABIL extends battery life in all 120 stress-test runs, and is ready for a 50-home pilot.



Working prototype: the family screen of the controller, replaying real Maiduguri weather hour by hour.

- Bayesian optimization of the rules (80 settings, tested on unseen 2019-2023 weather) found more battery life only by losing essential hours, so we keep the reliable rules.
- Top risk: the ageing model comes from lab cells. The pilot logs charge and temperature hourly to refit it on our packs.



Pilot

50 homes in Maiduguri, 6 months, with a DARES-registered solar company; logged data.

Expand

Borno, Yobe, Adamawa, where lines are attacked, via existing pay-per-use companies.

Scale

License the controller (firmware + $12 of sensors) to kit makers across the hot Sahel.

Our ask

A pilot partner in Borno and about $40,000 for 50 kits with data logging.

Stress tests: 96 Monte Carlo samples, 24 random-demand runs, 2 ageing models. Code and data: github.com/YounesToukal/SABIL. Pilot: 50 kits at $638 + ~$8,000 field work (estimate).